

Sumukh Kashi

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Education

University of Massachusetts Amherst

Expected May 2028

Master of Science in Computer Science

- **Relevant Coursework:** Systems for Data Science, Machine Learning, Neural Networks

BMS College of Engineering, VTU India

August 2024

Bachelor of Engineering in Information Science and Engineering

GPA – 8.95 /10

- **Relevant Coursework:** Object Oriented Programming, Operating Systems, Database Management Systems, Artificial Intelligence

Professional Experience

Honeywell Aerospace

Bengaluru, India

Software Engineer I

Oct 2024 – Jul 2026

- Built a retrieval augmented generation chatbot (RAG) using Mistral 7B (Ollama) for internal engineering documentation retrieval, saving ~\$120K annually in support overhead.
- Designed retrieval pipeline using document chunking, embedding generation, vector search and context-augmented prompt construction for a 30+ engineer team.
- Optimized real-time OS APIs for fault logging and designed a certified fast-file system for the Anthem avionics operating system.
- Delivered safety-critical avionics software compliant with DO-178C standards, supporting certification objectives for aerospace programs.
- Coordinated with systems, verification, and application engineering teams to support software integration, testing, and issue resolution across program milestones.
- Contributed to Agile execution through sprint planning, backlog management, code reviews, and defect triage using Jira and Bitbucket.

Systems Engineer Intern

Mar 2024– Aug 2024

- Developed user interface connectivity and datalink feature in flight management system for Airbus aircraft family, supporting 3 interface functions.
- Implemented a search automation tool to efficiently retrieve and analyze user requirement specifications, reducing average lookup time from 7 minutes to less than 2 minutes per requirement.

iMedrix Systems

Bengaluru, India

Machine Learning Engineer Intern

May 2023 – Aug 2023

- Developed a deep learning model to predict Blood Pressure from Photoplethysmography (PPG) waveforms.
- Reduced the model's Mean Absolute Error by ~33% and prediction variation by ~38% over the prior baseline demonstrating production viable signal modelling.

Projects

Carbon Footprint Prediction using Machine Learning and User Recommendation System using Reinforcement Learning

- Designed a dual-phase framework using ML (RMSE as low as 0.06 kgCO₂e) and reinforcement learning to generate recommendations, reducing emission variance by up to 10% for electrical appliances and transport vehicles.

Drone Detection System using YOLOv5.0

- Designed a drone detection system to detect air-borne and stationary drones using YOLOv5.0. Used a medium-dense layer having 391 layers to enhance mean average precision (mAP). Tuned hyperparameters to achieve 82% mAP@0.5 and 57% mAP@0.5:0.95 on 500 held-out test images.

Predictive Modeling of Electricity Generation from Fuel Cells

- Reconstructed time-series data for 3 variables - pH, TSS, and COD using a graph digitizer and developed a CNN-LSTM regressor to predict microbial fuel-cell voltage output, achieving RMSE of 0.04 V on test data.

Research Publications

- CNN-LSTM model for Electricity Prediction from microbial fuel cells- [IEEE NMITCON 2023 \(Best Paper\)](#)
- eGRUMET: Enhanced GRU for English-to-Kannada Neural Machine Translation - [IEEE ICCNT 2023 \(First Author\)](#)
- ConForMiSt: ML + RL for Carbon Footprint Optimization - [NEPT 2025 \(First Author\)](#)
- Enhancing Immersive User Experience Quality of StudoBot Telepresence Robots with Reinforcement Learning - [IEEE AICAPS 2023](#)
- A survey and analysis of drone detection systems using a systems approach superposed on scenarios - [Wiley Systems Engineering, 2024](#)

Awards & Recognition

- Dashing Debut Award – Aerospace Technologies – Honeywell Quarterly Awards Apr 2025
- US Patent filed (First Time Inventor Award) Dec 2025
- Meritorious Invention Disclosure Award - Aircraft Cockpit Display System Jan 2026
- Best Research Paper Award – IEEE NMITCON Sep 2023
- Runner-up – IdeaForge IIT Bombay Innovation Challenge Jul 2023

Skills

- **Languages:** Python, C++
- **Tools and Platforms:** Jira, Bitbucket, IBM DOORS/DNG
- **ML/DL:** PyTorch, TensorFlow, OpenCV, sci-kit learn, Hugging Face
- **GenAI:** RAG, embeddings, vector search, Ollama